IE 203 - Operations Research II

Midterm

İ. Kuban Altınel **April 08, 2026**

Duration: 1.5 hours
Important notice: 1. This is a CLOSED BOOK exam.
2. You may keep the questions.

Question I (30 pts.)

Consider the mixed integer nonlinear programming (MINLP) problem

$$\begin{array}{lrrl} \max & z = & -4x_1 - x_2 & \\ & \text{s.t.} & 2x_1 + 3x_2 & \geq 4 \\ & & 2x_1^2 + x_2 & \geq 2 \\ & & -x_1 + x_2 & \leq 3 \\ & & x_1, x_2 & \geq 0 \\ & & x_2 & \text{integer} \end{array}$$

a. Show graphically the feasible solution set.

b. Solve graphically the MINLP.

c. Show graphically the convex hull of the feasible solution set.

d. Find the extreme points and extreme directions of the convex hull of the feasible solution set.

e. Solve graphically the LP consisting of the minimization of the objective function subject to the constraints describing the convex hull. Is the optimal solution same as the one of the original MINLP obtained in (b)?

Question II (40 pts.)

Consider the mixed integer linear programming (MILP) problem

$$\begin{array}{lrrl} \text{MILP:} & z_{IP} = \max & 8x_1 + 5x_2 & \\ & \text{s.t.} & x_1 + x_2 & \leq 6 \\ & & 9x_1 + 5x_2 & \leq 45 \\ & & x_1, x_2 & \geq 0 \\ & & x_2 & \textit{integer}. \end{array}$$

a. Solve the LP relaxation of the problem using simplex algorithm and obtain optimal tableau.

b. Generate the Gomory cut corresponding to the basic variable violating integrality restriction.

c. Add the new cut to the obtained optimal tableau and reoptimize it using dual simplex algorithm for finding the next LP solution.

d. Show graphically feasible solution set with the generated Gomory cut.

Question III (30 pts.)

Consider the following unconstrained problem:

$$\min f(x_1, x_2) = x_1^2 - x_1x_2 + 2x_2^2 - 2x_1 + e^{x_1+x_2}$$

a. Write the first-order necessary optimality conditions. Are these conditions also sufficient for optimality? Why?

b. Is $\mathbf{x}^* = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ an optimal solution? If not, identify the steepest descent direction at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and the step length using exact line search along the steepest descent direction.

c. Repeat (b) for Newton's method. You do not have to calculate the step length. It is enough to give the function in α, which is to be minimized.

d. Solve this problem after dropping the last term in the objective function.

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Midterm - Solutions

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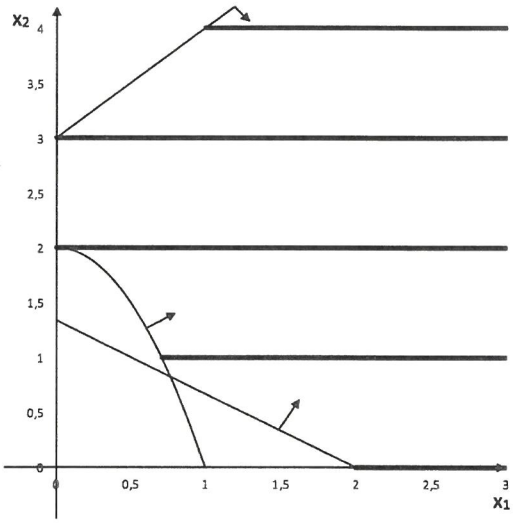
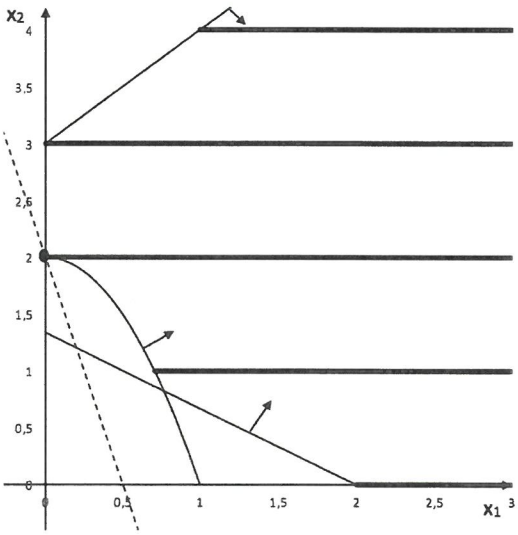
Question I (30 pts.)

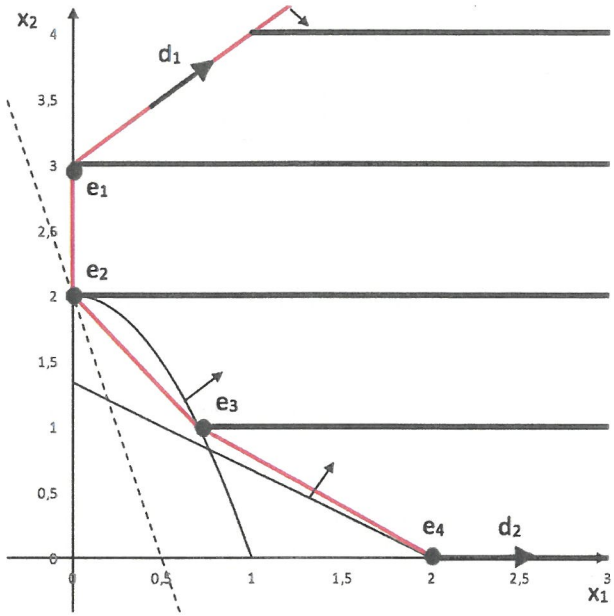
a. The feasible solution set is given with the thick lines on the following figure:



b. One can see that the optimum solution is $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ with objective function value $\mathbf{z}^* = -2$.



c. The boundaries of the convex hull is given with the red lines on the following figure:



d. As can be seen from the figure given in part (c) the extreme points are $\mathbf{e_1} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \mathbf{e_2} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \mathbf{e_3} = \begin{pmatrix} 1/\sqrt{2} \\ 1 \end{pmatrix}, \mathbf{e_4} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and the extreme directions are $\mathbf{d_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{d_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

e. Minimizing the same objective function over the convex hull of the original feasible region will give unbounded optimal value, since the convex hull is unbounded.

Question II (40 pts.)

$$\begin{array}{llrl} \text{MILP:} & z_{IP} = \max & 8x_1 + 5x_2 & \\ & \text{s.t.} & x_1 + x_2 & \leq 6 \\ & & 9x_1 + 5x_2 & \leq 45 \\ & & x_1, x_2 & \geq 0 \\ & & x_2 & \textit{integer}. \end{array}$$

a. Optimal tableau:

	z	x_1	x_2	x_3	x_4	rhs
z	1	0	0	1.25	0.75	41.25
x_2	0	0	1	2.25	−0.25	2.25
x_1	0	1	0	−1.25	0.25	3.75

Here, x_3 and x_4 are respectively the slack variables of the first and second inequalities.

b. In the optimal solution $x_1 = 3.75$ and $x_2 = 2.25$. Since x_2 is restricted to be integer, Gomory cut must be based on x_2 and the corresponding row of the optimal tableau gives

$$(1+0)x_2 + (2+0.25)x_3 + (-1+0.75)x_4 = 2 + 0.25$$

from which Gomory cut

$$x_2 + 2x_3 - x_4 \leq 2$$

or equivalently

$$-0.25x_3 - 0.75x_4 \leq -0.25$$

can be obtained.

c. New Gomory cut can be added to the optimal tableau given in (a) as the last row with its slack variable x_5 being the new basic variable. This operation results in the following tableau.

	z	x_1	x_2	x_3	x_4	x_5	rhs
z	1	0	0	1.25	0.75	0	41.25
x_2	0	0	1	2.25	−0.25	0	2.25
x_1	0	1	0	−1.25	0.25	0	3.75
x_5	0	0	1	2	−1	1	2

This tableau is not appropriate because of the nonzero entry, i.e., 1, under the column of x_2 in the last row. This can be fixed by subtracting the second row from the last one, which makes the tableau

	z	x_1	x_2	x_3	x_4	x_5	rhs
z	1	0	0	1.25	0.75	0	41.25
x_2	0	0	1	2.25	−0.25	0	2.25
x_1	0	1	0	−1.25	0.25	0	3.75
x_5	0	0	0	−0.25	−0.75	1	−0.25

As can be noticed a permutation matrix occurs in the columns of the basic variables in rows 2, 3, and 4. The new tableau is appropriate but primal infeasible: x_5 violates nonnegativity restriction. The new optimal optimal tableau for the LP relaxation is given below. It is obtained using dual simplex algorithm.

	z	x_1	x_2	x_3	x_4	x_5	rhs
z	1	0	0	1	0	1	41
x_2	0	0	1	7/3	0	−1/3	7/3
x_1	0	1	0	−4/3	0	1/3	11/3
x_4	0	0	0	1/3	1	−4/3	1/3

d. Gomory cut $x_2+2x_3-x_4 \leq 2$ can be equivalently expressed in x_1 and x_2 after replacing $x_3 = 6-x_1-x_2$ and $x_4 = 45 - 5x_1 - 9x_2$ as

$$7x_1 + 4x_2 \leq 35,$$

which cuts previous relaxed optimal solution $x_1 = 3.75$, $x_2 = 2.25$ since $7\times 3.75+4\times 2.25 = 35.25 > 35$. Obviously the new relaxed optimal is not an optimal solution of the MILP since $x_2 = 7/3$ violates the integrality restriction on x_2.

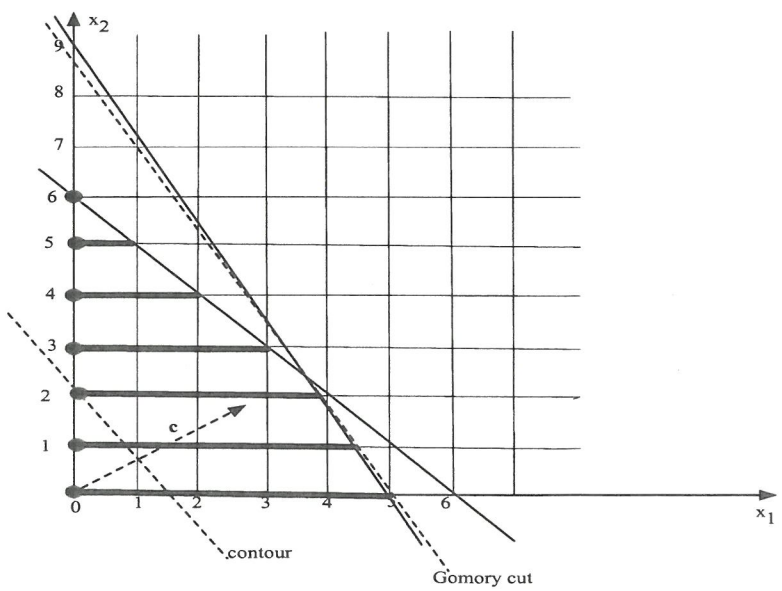


Figure 1: Gomory cut

Question III (30 pts.)

a. The first order necessary conditions imply

$$\nabla f(x_1, x_2) = 0 \Longrightarrow \begin{cases} 2x_1 - x_2 + e^{x_1+x_2} = 2 \\ -x_1 + 4x_2 + e^{x_1+x_2} = 0 \end{cases}$$

The Hessian matrix is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} 2 + e^{x_1+x_2} & e^{x_1+x_2} - 1 \\ e^{x_1+x_2} - 1 & 4 + e^{x_1+x_2} \end{pmatrix}.$$

It is positive definite over the domain of the function since the leading principal minors are $2 + e^{x_1+x_2}$ and $7 + 8e^{x_1+x_2}$, which are positive for all x_1 and x_2. Therefore, the first order necessary conditions are also sufficient. In fact the function is strictly convex.

b. $x^* = \begin{pmatrix} 0 & 0 \end{pmatrix}^T$ is not an optimal solution. $\nabla f(0,0) = \begin{pmatrix} -1 & 1 \end{pmatrix}$ and any direction vector $\mathbf{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$ such that $-d_1 + d_2 < 0$ is a descent direction of $f(x_1, x_2)$ at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Let us select $\mathbf{d} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, which is $-\nabla^T f(0,0)$ namely the steepest descent direction.

Then, $f(\mathbf{x} + \alpha\mathbf{d}) = \alpha^2 + \alpha^2 + 2\alpha^2 - 2\alpha = g(\alpha) = 4\alpha^2 - 2\alpha$, $g'(\alpha) = 8\alpha - 2 = 0$ and $\alpha^* = \frac{1}{4}$. Besides, $g''(\alpha) = 8 > 0$. Hence, $\alpha^* = \frac{1}{4}$ minimizes $g(\alpha)$ and can be chosen as the step length.

c. The Hessian matrix at $x^* = \begin{pmatrix} 0 & 0 \end{pmatrix}^T$ is $\begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$ with inverse $\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$. Hence, the Newton direction is $\mathbf{d} = -\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{5} \end{pmatrix}$.

Then, for pure Newton's method $\alpha = 1$. However, for Newton's method with exact line search $f(\mathbf{x} + \alpha\mathbf{d}) = f(\frac{\alpha}{3}, -\frac{\alpha}{5}) = h(\alpha) = \frac{58}{225}\alpha^2 - \frac{2}{3}\alpha + \exp(\frac{2}{15}\alpha)$. Notice that $h''(\alpha) = \frac{116}{225} + \frac{4}{225}\exp(\frac{2}{15}\alpha) > 0$ and thus $h(\alpha)$ is strictly convex. As a consequence the step length is the solution of $h'(\alpha) = 0$, namely α such that $58\alpha + 15\exp(\frac{2}{15}\alpha) = 75$.

d. The function to be considered is $f(x_1, x_2) = x_1^2 - x_1x_2 + 2x_2^2 - 2x_1$ and the first order necessary conditions imply

$$\begin{cases} 2x_1 - x_2 = 2 \\ -x_1 + 4x_2 = 0 \end{cases}$$

from which

$$x_1^* = \frac{8}{7}$$
$$x_2^* = \frac{2}{7}$$

follows. It is the unique stationary point.

The Hessian matrix is $\begin{pmatrix} 2 & -1 \\ -1 & 4 \end{pmatrix}$ with leading principal minors 2 and 7, which makes it positive definite. Thus, the function is strictly convex and $\begin{pmatrix} \frac{8}{7} \\ \frac{2}{7} \end{pmatrix}$ is the unique global minimum. The optimal value of the function is $f(x^*) = -\frac{56}{49}$.